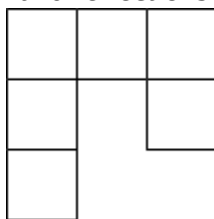


IMO(45th) – 2004

First Day

- Let ABC be an acute angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote O the midpoint of the side BC . The bisectors of $\angle BAC$ and $\angle MON$ intersect at R . Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC .
- Find all polynomials f with real coefficients such that for all reals a, b, c such that $ab + bc + ca = 0$ we have the following relations $f(a - b) + f(b - c) + f(c - a) = 2f(a + b + c)$.
- Define a “hook” to be a figure made up of six unit squares as shown below in the picture, or any of the figures obtained by applying rotation and reflections to this figure.



Determine all $m \times n$ rectangles that can be covered without gaps and without overlaps with hooks such that :

- The rectangle is covered without gaps and without overlaps.
- No part of a hook covers area outside the rectangle.

Second Day

- Let $n \geq 3$ be an integer. Let $t_1, t_2, \dots, t_n$ be positive numbers such that $n^2 + 1 > (t_1 + t_2 + \dots + t_n) \left(\frac{1}{t_1} + \frac{1}{t_2} + \dots + \frac{1}{t_n} \right)$. Show that t_i, t_j, t_k are sides lengths of a triangle for all i, j, k with $1 \leq i < j < k \leq n$.
- In a convex quadrilateral $ABCD$ the diagonal BD does not bisect the angles ABC and CDA . The point P lies inside $ABCD$ and satisfies $\angle PBC = \angle DBA$, $\angle PDC = \angle BDA$. Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.
- We call a positive integer alternating if every two consecutive digits in its decimal representation are of different parity. Find all positive integers n has a multiple which is alternating.

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